ECE253: A UNIFIED COMPARATIVE FRAMEWORK FOR CLASSICAL AND LEARNING-BASED IMAGE RESTORATION: DEHAZING, DEBLURRING, AND DENOISING

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ABSTRACT

This project investigates practical single-image restoration for campus-facing imaging conditions by building a unified evaluation framework across three tasks: dehazing, deblurring, and denoising. We benchmark classical, prior-driven pipelines against learning-based models using a consistent protocol on both public datasets and real-world captures. For dehazing, we compare a physics-based Dark Channel Prior (DCP) method and a lightweight deep baseline (AOD-Net) on synthetic haze with paired ground truth as well as real-haze data and self-captured outdoor scenes, reporting full-reference metrics when available and no-reference perceptual metrics otherwise. For deblurring, we evaluate the L0-based blind deblurring approach of Xu *et al.* and the internal patch recurrence method of Michaeli *et al.* on a standard motion-blur benchmark and real blurred photos, analyzing both fidelity and perceptual naturalness. For denoising, we compare multi-wavelet shrinkage, support vector regression (SVR), and a residual CNN (DnCNN) under controlled Gaussian noise levels and on real noisy images. Across all tasks, the experiments reveal consistent and interpretable trade-offs. Classical methods remain competitive and often generalize reliably under domain shift or limited training data, but they are sensitive to hyperparameters and may introduce characteristic artifacts (e.g., haloing in dehazing or oversharpening/texture loss in deblurring). Learning-based models achieve stronger restoration quality when the test distribution is aligned with training and sufficient data are available, providing improved structure preservation and more stable perceptual quality. In particular, residual deep denoising demonstrates strong robustness under heavy corruption, indicating the benefit of learned convolutional priors for separating noise from image content.

1 Introduction

1.1 Background and Motivation

Image restoration is a fundamental problem in digital image processing, aiming to recover high-quality visual information from degraded observations. It underpins a wide range of real-world applications, including autonomous driving, aerial inspection, medical imaging, surveillance, and consumer photography. In practical imaging systems, however, captured images are often corrupted by multiple types of degradations, which significantly reduce visual quality and limit downstream usability.

Unlike idealized laboratory conditions, real-world image acquisition is subject to both environmental and hardware constraints. Outdoor scenes are affected by atmospheric scattering and weather conditions, handheld devices suffer from camera shake and motion, and low-cost sensors introduce noticeable noise under limited lighting. These constraints motivate the need for robust image restoration methods that can operate reliably under heterogeneous and realistic degradation settings.

1.2 Problem Statement: Imaging Degradations in Campus Scenarios

A university campus provides a representative real-world imaging environment where multiple degradation types naturally arise. Images are routinely captured for lecture recording, laboratory documentation, project reports, campus surveillance, and student-created media. In this context, three common degradation modes are particularly prevalent:

Dehazing: Outdoor campus scenes often suffer from haze and reduced contrast due to atmospheric scattering caused by humidity, fog, or pollution. This degradation primarily affects long-range visibility and color fidelity, making scene understanding and visual inspection more difficult.

Deblurring: Images captured using handheld or mobile devices are frequently degraded by motion blur, especially under low-light conditions or dynamic acquisition scenarios. Camera shake and object motion introduce unknown blur kernels, leading to loss of structural details and edge sharpness.

Denoising: Indoor environments such as classrooms and laboratories often rely on low-cost imaging sensors operating at high ISO settings. As a result, captured images exhibit noticeable noise, which degrades texture details and overall visual clarity.

These degradations significantly reduce the usability of campus images for academic, educational, and monitoring purposes, motivating a unified image restoration framework tailored to realistic acquisition conditions.

1.3 Related Work: Classical and Learning-Based Methods

1.3.1 Dehazing

Image dehazing is typically formulated under the atmospheric scattering model, where the observed hazy image is expressed as a mixture of the true scene radiance attenuated by the medium transmission and an additive airlight component determined by the global atmospheric light [1].

Classical methods. He *et al.* introduced the Dark Channel Prior (DCP), which estimates the transmission map from local minimum statistics and recovers the clean image using guided filtering [2]. This method is simple, interpretable, and grounded in physical imaging assumptions, and it performs well on outdoor scenes with moderate haze. Subsequent works improved DCP by introducing better sky-region handling and boundary constraints to alleviate halo artifacts and color distortion using edge-preserving filters and alternative priors [3].

Learning-based methods. Learning-based dehazing approaches employ convolutional neural networks to estimate either the transmission map or the clean image directly, often achieving higher PSNR and SSIM on benchmark datasets [4, 5]. However, these methods typically require paired hazy/clear training data and may exhibit limited generalization when deployed in real-world environments that differ from the training distribution, especially under unseen haze conditions [6].

1.3.2 Deblurring

Image deblurring aims to recover a sharp latent image from a blurred observation, where blur is commonly caused by camera shake, object motion, or defocus.

Classical image deblurring methods typically assume a known blur kernel and invert the convolution process to get the original image. Wiener filtering performs the inverse convolution operation in the frequency domain and suppresses the amplification of high-frequency noise through the noise regularization term. It performs well when the kernel are known accurately, but suffers from over smoothing and ringing artifacts under kernel mismatch. Richardson–Lucy deconvolution iteratively solves through an expectation maximization framework based on Poisson statistics to ensure the non-negativity of the reconstructed image. However, RL is sensitive to noise and often amplifies artifacts when iteration too many times. [7]

Blind deblurring algorithms estimate both the unknown blur kernel and the latent sharp image. Levin et al. (2009)[8] proposed a Bayesian framework that maximizes the marginal likelihood over blur kernels instead of jointly optimizing the image and kernel. This approach effectively avoids the “no-blur” degeneracy that appears in maximum a posteriori (MAP) estimation, which shows more stable kernel recovery. Cho and Lee (2009)[9] introduced a fast edge based blind deblurring method, which uses bilateral filters and impulse filters to detect and enhance strong edges to effectively guide kernel updates. Although both methods improved the robustness and speed of blind deconvolution, they still limited in handling spatially variant blur and are sensitive to noise and textureless regions.

Deep learning approaches have reached advanced deblurring performance by training from large datasets. CNN and Transformer-based models such as DeblurGAN, MPRNet, and Restormer achieve higher restoration quality. However, they rely on synthetic training data and often restricts generalization to real world dataset. [10]

This project reproduces and evaluates two methods: (1) Blind Deblurring Using Internal Patch Recurrence (2) Unnatural L_0 Sparse Representation for Natural Image Deblurring. The project aims to provide a systematic comparison of their performance, robustness, and computational efficiency on standard benchmark datasets and real-world blurred images.

1.3.3 Denoising

Image denoising aims to recover a clean image from noisy observations, where noise typically arises from sensor limitations and high ISO settings.

Classical methods. Early denoising methods rely on wavelet shrinkage or variational regularization, where small wavelet coefficients dominated by noise are attenuated and large coefficients corresponding to edges and textures are preserved [11]. To better handle spatially varying smoothness, projecting onto multiple wavelet or Besov balls via projection onto convex sets (POCS) can further improve PSNR by approximately 1–2 dB compared to single-basis shrinkage [12].

Learning-based methods. Learning-based denoising approaches formulate denoising as a supervised regression problem. Li *et al.* employed support vector regression (SVR) with an RBF kernel, demonstrating improved performance over purely wavelet-based methods on certain non-natural images [13]. More recently, DnCNN introduced residual learning and batch normalization to directly predict the noise component under additive white Gaussian noise (AWGN), achieving strong performance on standard benchmarks [14]. Subsequent works further enhance perceptual consistency by adopting deeper architectures or quality-aware training strategies [15].

1.4 Research Gaps

Despite extensive progress, several limitations remain in existing studies. First, many works focus on a single restoration task in isolation, making it difficult to understand whether performance gains stem from the method itself or from task-specific assumptions. Second, comparisons between classical and learning-based approaches are often performed under inconsistent datasets and evaluation protocols. Third, a significant portion of learning-based methods are trained and tested on synthetic data, which may not accurately reflect real-world degradation patterns encountered in practical imaging scenarios.

These factors motivate a unified evaluation framework that enables fair and systematic comparison across different restoration tasks and methodological paradigms.

1.5 Project Overview

In this project, we conduct a unified comparative study of classical and learning-based image restoration methods across three representative tasks:

- **Direction A (Dehazing):** comparison between a physics-based Dark Channel Prior (DCP) method and a lightweight ResNet-based dehazing network.
- **Direction B (Deblurring):** comparison between two blind deblurring methods based on sparse gradients and internal patch recurrence.
- **Direction C (Denoising):** comparison among a classical multi-wavelet/Besov projection method, an example-based SVR denoiser, and a modern residual CNN denoiser (DnCNN).

All methods are implemented and evaluated under consistent experimental settings using PSNR, SSIM, and NIQE as evaluation metrics. Both public benchmarks (e.g., RESIDE, BSD68) and self-collected campus images are used to assess quantitative performance and real-world generalization.

1.6 Contributions

The main contributions of this project are summarized as follows:

- A unified experimental framework for comparing classical and learning-based image restoration methods across dehazing, deblurring, and denoising tasks.

- A systematic evaluation using consistent metrics (PSNR, SSIM, NIQE) on both public benchmarks and real-world campus images.
- A task-level analysis of the strengths and limitations of analytical priors versus learned representations under realistic acquisition conditions.

2 Method

2.1 Direction A: Image Dehazing

2.1.1 Dark Channel Prior (DCP)

Image dehazing is commonly formulated using the atmospheric scattering model. Given a hazy observation I, the goal is to recover the haze-free radiance J. The formation model is:

$$I(x) = J(x)t(x) + A(1 - t(x)), \tag{1}$$

where x denotes the pixel location, A is the global atmospheric light, and $t(x) \in [0, 1]$ is the transmission map describing the portion of scene radiance that reaches the camera.

Our classical baseline, the Dark Channel Prior (DCP), estimates transmission based on the observation that in most non-sky patches of haze-free outdoor images, at least one color channel has very low intensity. We define the dark channel of an image J as:

$$J^{\text{dark}}(x) = \min_{c\in\{r,g,b\}} \left(\min_{y\in\Omega(x)} J^c(y) \right), \tag{2}$$

where $\Omega(x)$ is a local window centered at x. Under hazy conditions, the observed dark channel becomes brighter. Assuming A is known (estimated from the brightest pixels in the dark channel), the coarse transmission $\tilde{t}(x)$ is estimated as:

$$\tilde{t}(x) = 1 - \omega \min_c \left(\min_{y\in\Omega(x)} \frac{I^c(y)}{A^c} \right), \tag{3}$$

where $\omega \in (0, 1]$ controls the amount of haze removed and is tuned in our experiments.

However, the patch-based transmission estimate $\tilde{t}(x)$ can be noisy due to local minimum operations. In our implementation, we keep the baseline pipeline lightweight and deterministic: we directly apply a lower bound t_0 to prevent noise amplification in heavy haze regions, and recover the scene radiance by:

$$J(x) = \frac{I(x) - A}{\max(t(x), t_0)} + A. \tag{4}$$

We further perform a simple parameter search over the patch size, ω, and t_0 on the synthetic benchmark to identify a stable configuration for our evaluation.

Algorithm 1 Dark Channel Prior (DCP) Dehazing (Implementation Used)

Require: Hazy image I, patch size sz, parameters ω, t_0
Ensure: Dehazed image J
1: Convert I to floating point and normalize to $[0, 1]$.
2: Compute dark channel: $D(x) \leftarrow \min_{c\in\{r,g,b\}} \min_{y\in\Omega_{sz}(x)} I^c(y)$.
3: Estimate atmospheric light A from high-intensity regions (selected using D), then set $A \in R^3$.
4: Estimate transmission: $t(x) \leftarrow 1 - \omega \min_c \min_{y\in\Omega_{sz}(x)} \frac{I^c(y)}{A^c}$.
5: Apply lower bound to avoid amplification: $t(x) \leftarrow \max(t(x), t_0)$.
6: Recover radiance: $J(x) \leftarrow \frac{I(x)-A}{t(x)} + A$.
7: Clip J to $[0, 1]$ and convert to uint8 image.
8: **return** J

2.1.2 AOD-Net: Learning-Based End-to-End Dehazing

To compare the physics-based DCP with a modern data-driven approach, we employ the All-in-One Dehazing Network (AOD-Net). Unlike traditional methods that estimate $t(x)$ and A separately, AOD-Net learns a direct mapping from the hazy input I to the restored output J using a lightweight Convolutional Neural Network (CNN).

The method reformulates the atmospheric scattering model into a unified representation. By introducing a new variable $K(x)$ defined as:

$$K(x) = \frac{\frac{1}{t(x)}(I(x) - A) + (A - b)}{I(x) - b}, \tag{5}$$

the physical model can be rewritten as an end-to-end affine transformation:

$$J(x) = K(x) \odot I(x) - K(x) + b, \tag{6}$$

where $\odot$ denotes element-wise multiplication and b is a constant bias (set to 1). This formulation allows the network to implicitly learn both the transmission and atmospheric light within a single parameter $K(x)$, minimizing the accumulation of reconstruction errors.

In our experiment, we utilize a pre-trained AOD-Net model as an off-the-shelf baseline. The inference pipeline is consistent across datasets: input images are normalized to $[0, 1]$, forwarded through the network to produce the dehazed output directly, and then clipped to the valid intensity range. No additional fine-tuning is performed.

Algorithm 2 AOD-Net Dehazing Inference (Implementation Used)

Require: Hazy image I, pre-trained model f_θ
Ensure: Dehazed image J
1: Convert I to floating point and normalize to $[0, 1]$.
2: Reorder to model tensor format and run inference: $J \leftarrow f_\theta(I)$.
3: Clip J to valid range $[0, 1]$.
4: Convert J to uint8 image and save/output.
5: **return** J

2.2 Direction B: Image Deblurring

2.2.1 Unnatural L0 Sparse Representation for Natural Image Deblurring

The first algorithm we reproduce is the L0-based blind deblurring framework of Xu et al.[16] This method is based on the idea that natural images have high gradient sparsity and that applying an approximate L_0penalty to image gradients creates an "unnatural representation" consisting only of salient structural edges. This representation stabilizes kernel estimation and prevents the degeneracy commonly observed in maximum a posteriori (MAP)-based blind deconvolution.

The degradation model assumes

$$y = \sum_m k_m H_m x + \varepsilon, \tag{7}$$

where H_m denotes spatial transformations corresponding to camera motion. The joint optimization over the latent image $\tilde{x}$ and blur kernel k is formulated as

$$\min_{\tilde{x},k} \left\| \sum_m k_m H_m \tilde{x} - y \right\|_2^2 + \lambda \sum_{*\in\{h,v\}} \phi_0(\partial_* \tilde{x}) + \gamma \|k\|_2^2. \tag{8}$$

The sparsity term $\phi_0(\cdot)$ approximates the L_0 gradient count through the piecewise function

$$\phi(\partial z_i) = \begin{cases} \frac{1}{\epsilon^2}|\partial z_i|^2, & |\partial z_i| \le \epsilon, \\ 1, & \text{otherwise}, \end{cases} \tag{9}$$

which retains sharp edges while suppressing low-amplitude structures.

Optimization proceeds by alternating between solving for $\tilde{x}$ and updating k. The latent image update admits a closed-form solution in the Fourier domain, while kernel estimation employs a quadratic form that also benefits from FFT acceleration. A coarse-to-fine pyramid together with a Graduated Non-Convexity schedule gradually tightens the sparsity constraint by decreasing ϵ, thus ensuring both stability and rapid convergence. After kernel estimation converges, a non-blind deconvolution stage using a hyper-Laplacian prior reconstructs the final natural-looking deblurred image.

Algorithm 3 L0-based Blind Deblurring (Simplified)

Require: Blurry image y
Ensure: Estimated kernel $\hat{k}$ and restored image x
Build an image pyramid $\{y^{(s)}\}$ from coarse to fine.
Initialize kernel k at the coarsest level.
for each pyramid level s (coarse $\to$ fine) **do**
    Initialize latent image $\tilde{x} \leftarrow y^{(s)}$.
    for a fixed number of iterations **do**
        Compute image gradients $\partial_h \tilde{x}$, $\partial_v \tilde{x}$.
        Apply L0-like gradient thresholding to obtain sparse gradients.
        Update $\tilde{x}$ by minimizing the data term

$$\left\| k * \tilde{x} - y^{(s)} \right\|_2^2$$

        plus the sparse regularizer, using FFT.
        Update kernel k by solving a quadratic problem with

$$\left\| k * \tilde{x} - y^{(s)} \right\|_2^2 + \gamma \|k\|_2^2,$$

        and normalize k.
    end for
end for
Set $\hat{k} \leftarrow k$ at the finest level.
Perform final non-blind deconvolution with $\hat{k}$ to obtain x.
return $(\hat{k}, x)$

2.2.2 Blind Deblurring Using Internal Patch Recurrence

The second method is based on the internal patch recurrence prior introduced by Michaeli et al.[17] A key empirical observation is that small patches in a sharp natural image tend to recur across scales: when an image is downsampled by a factor α, most patches reappear nearly unchanged. Blur disrupts this phenomenon, as blurry patches fail to find close counterparts at coarser scales. Thus, the optimal blur kernel is the one whose removal maximizes cross-scale patch similarity.

The method estimates the latent image x and kernel k by solving

$$\min_{x,k} \|y - k * x\|_2^2 + \lambda_1 \rho(x, x_\alpha) + \lambda_2 \|k\|_2^2, \tag{10}$$

where x_α is the downscaled version of x, and the prior term $\rho(x, x_\alpha)$ evaluates the dissimilarity between patches of x and their nearest neighbors in x_α. Using nonparametric kernel density estimation, the algorithm constructs a patch-based prior directly from the image itself, thus avoiding reliance on external datasets. Since downscaling inherently sharpens image structures by narrowing the effective blur, the patches in x_α provide examples of how patches in x should appear after partial blur removal.

Each iteration alternates between three components: generating x_α using sinc-based downscaling; updating x using patch matching followed by solving a linear system in the Fourier domain; and refining k through constrained least-squares. The entire optimization is executed over a pyramid from coarse to fine, improving stability and preventing divergence. Compared to gradient-based priors, this approach excels in scenes with rich textures or weak edge information, where traditional sparsity assumptions fail.

Algorithm 4 Blind Deblurring Using Internal Patch Recurrence (Simplified)

Require: Blurry image y
Ensure: Estimated kernel $\hat{k}$ and restored image x
Build an image pyramid $\{y^{(s)}\}$ from coarse to fine.
At the coarsest level, set $k \leftarrow \delta$, $x \leftarrow y^{(1)}$.
for each pyramid level s (coarse $\to$ fine) **do**
    if $s > 1$ **then**
        Upsample x and k from level $s-1$ to level s.
    end if
    for a fixed number of iterations **do**
        Downscale current estimate x to x_α using sinc downsampling.
        For each patch in x, find its nearest patch in x_α and build a proxy image z.
        Update x by solving

$$\min_x \|y^{(s)} - k * x\|_2^2 + \lambda_1 \|x - z\|_2^2$$

        in the Fourier domain.
        Update kernel k by solving

$$\min_k \|y^{(s)} - k * x\|_2^2 + \lambda_2 \|k\|_2^2,$$

        then enforce non-negativity and normalization.
    end for
end for
Set $\hat{k} \leftarrow k$ at the finest level.
Optionally refine x via a final non-blind deconvolution with $\hat{k}$.
return $(\hat{k}, x)$

2.3 Direction C: Image Denoising

2.3.1 Multi-Wavelet Shrinkage for Image Denoising

The first denoising algorithm we implement is a classical multi-wavelet shrinkage method based on sparse representations of natural images. The key assumption is that clean images admit sparse coefficients in the wavelet domain, while additive white Gaussian noise remains approximately i.i.d. across coefficients. By enforcing sparsity through coefficient shrinkage, noise can be effectively suppressed while preserving edges and structural details.

Under the additive noise model

$$\mathbf{y} = \mathbf{x} + \mathbf{n}, \qquad \mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \tag{11}$$

wavelet-based denoising can be interpreted as solving a regularized least-squares problem

$$\min_{\mathbf{x}} \frac{1}{2}\|\mathbf{y} - \mathbf{x}\|_2^2 + \lambda \|\mathbf{W}\mathbf{x}\|_1, \tag{12}$$

where $\mathbf{W}$ denotes a discrete wavelet transform (DWT) and the ℓ_1 penalty promotes sparsity of wavelet coefficients. The corresponding solution is given by soft-thresholding

$$\mathcal{S}_\lambda(w_i) = \operatorname{sign}(w_i) \max(|w_i| - \lambda, 0). \tag{13}$$

While single-wavelet shrinkage performs well on homogeneous regions, it may fail to adapt to spatially varying textures and structures. To address this limitation, we adopt a multi-wavelet framework that combines several orthogonal wavelet bases $\{\mathbf{W}_m\}_{m=1}^M$. Each wavelet basis defines a convex constraint set

$$\mathcal{B}_m = \{\mathbf{x} : \|\mathbf{W}_m \mathbf{x}\|_1 \le \tau_m\}, \tag{14}$$

which can be interpreted as a Besov-norm constraint associated with that basis.

For the image output, the denoised image is obtained using a Projection Onto Convex Sets (POCS) iteration, implemented via wavelet-domain soft-thresholding followed by inverse DWT.

In our implementation, three Daubechies wavelet bases (db2, db4, sym8) are used. The threshold parameters are chosen using the universal rule

Algorithm 5 Multi-Wavelet Shrinkage via POCS (Simplified)

Require: Noisy image $\mathbf{y}$, noise variance σ^2
Ensure: Denoised image $\hat{\mathbf{x}}$
1: Initialize $\mathbf{x}^{(0)} \leftarrow \mathbf{y}$
2: Select wavelet bases $\{\mathbf{W}_m\}_{m=1}^M$
3: **for** $k = 0$ to $K-1$ **do**
4: **for** each wavelet basis $\mathbf{W}_m$ **do**
5: Compute wavelet coefficients $\mathbf{w}_m \leftarrow \mathbf{W}_m \mathbf{x}^{(k)}$
6: Apply soft-thresholding

$$\mathbf{w}_m \leftarrow \mathcal{S}_{\lambda_m}(\mathbf{w}_m)$$

7: Reconstruct image via inverse DWT $\mathbf{x}^{(k)} \leftarrow \mathbf{W}_m^{-1} \mathbf{w}_m$
8: **end for**
9: **end for**
10: Set $\hat{\mathbf{x}} \leftarrow \mathbf{x}^{(K)}$
11: **return** $\hat{\mathbf{x}}$

2.3.2 Support Vector Regression for Image Denoising

The second denoising algorithm we implement is a patch-based Support Vector Regression (SVR) method, which formulates image denoising as a supervised nonlinear regression problem. Instead of explicitly imposing analytical priors such as sparsity or smoothness, SVR learns a mapping from noisy local image patches to clean pixel intensities using labeled examples. This approach represents an intermediate paradigm between classical model-based denoisers and modern deep learning methods.

Under the additive noise model, the denoising task is reformulated at the patch level. For each pixel location, a local noisy patch centered at that pixel is extracted and vectorized as $\mathbf{x}_i$, while the corresponding clean center pixel intensity y_i serves as the regression target.

The SVR model seeks a nonlinear regression function of the form

$$f(\mathbf{x}) = \mathbf{w}^\top \phi(\mathbf{x}) + b, \tag{15}$$

where $\phi(\cdot)$ denotes an implicit feature mapping induced by a radial basis function (RBF) kernel. Model parameters are obtained by minimizing a regularized ε-insensitive loss, which balances prediction accuracy and model complexity. This formulation is known to provide robustness against noise while preventing overfitting when training data are limited.

In the training stage, 10000 noisy–clean patch pairs are set for randomly sampling from reference images. Each noisy patch is used as an input feature, and the corresponding clean center pixel is used as the regression label. During inference, the trained SVR model is applied in a sliding-window fashion: for each pixel in the noisy image, a local patch is extracted and fed into the regressor to predict the denoised intensity. Overlapping predictions are implicitly averaged through the dense scanning process.

Algorithm 6 SVR-based Denoising

Input: Noisy feature vectors $\{\mathbf{x}_i\}_{i=1}^N$
Output: Denoised signal $\hat{\mathbf{y}}$
Initialize SVR parameters (C, ε, γ) Train SVR model using RBF kernel Predict clean signal $\hat{\mathbf{y}} = f(\mathbf{x})$

2.3.3 Deep Residual CNN for Image Denoising (DnCNN)

The third denoising method adopted in this work is the Deep Denoising Convolutional Neural Network (DnCNN), which represents a modern learning-based approach for Gaussian image denoising. Unlike classical denoisers that rely on explicit analytical priors, DnCNN learns an implicit image prior directly from data using a deep convolutional architecture. The key design of DnCNN lies in its residual learning formulation combined with batch normalization, which enables stable training and strong generalization across noise levels.

Given a noisy observation, DnCNN is trained to predict the noise component rather than the clean image itself. Specifically, the network learns a residual mapping

$$\hat{\mathbf{n}} = F(\mathbf{y}; \Theta), \tag{16}$$

where $F(\cdot)$ denotes the CNN parameterized by Θ. The denoised image is then obtained by a simple subtraction. This residual formulation is motivated by the observation that the noisy input is often closer to the clean image than to the noise itself, making the residual mapping easier to optimize than a direct input–output mapping.

Network Architecture. The DnCNN architecture is derived from the VGG-style design and consists of a stack of convolutional layers without pooling. All convolutional filters have a spatial size of 3×3, and zero padding is applied to preserve spatial resolution throughout the network. For grayscale denoising, the network contains 17 layers in total: the first layer uses a Conv+ReLU block to extract low-level features, the intermediate layers employ Conv+BatchNorm+ReLU blocks to progressively separate image structures from noise, and the final layer uses a single convolution to reconstruct the residual noise image. With this depth, the receptive field reaches 35×35, which has been shown to provide a favorable trade-off between denoising performance and computational efficiency.

Training Objective. The network parameters are learned by minimizing the mean squared error between the predicted and ground-truth residuals over a training set $\{(\mathbf{y}_i, \mathbf{x}_i)\}_{i=1}^{N}$:

$$\mathcal{L}(\Theta) = \frac{1}{N} \sum_{i=1}^{N} \left\| F(\mathbf{y}_i; \Theta) - (\mathbf{y}_i - \mathbf{x}_i) \right\|_2^2. \tag{17}$$

Batch normalization is incorporated into the intermediate layers to reduce internal covariate shift and accelerate convergence. In our implementation, an additional structural similarity (SSIM) term is optionally included to improve perceptual quality under high noise levels.

Implementation Details. A pre-trained grayscale DnCNN model is fine-tuned on the BSD68 dataset under additive Gaussian noise levels $\sigma \in \{15, 25, 50\}$. Training is conducted using 40×40 image patches with a mini-batch size of 64. The Adam optimizer is employed with a learning rate of 10^{-4}, and fine-tuning is performed for three epochs to adapt the model while avoiding overfitting. During inference, batch normalization layers are fused into convolution weights to reduce computational overhead.

Algorithm 7 DnCNN-based Image Denoising

Input: Noisy image $\mathbf{y}$
Output: Denoised image $\hat{\mathbf{x}}$
Initialize CNN parameters Θ **foreach** *training sample* $(\mathbf{y}_i, \mathbf{x}_i)$ **do**
  $\mathbf{f}_i^{(1)} \leftarrow$ Conv+ReLU$(\mathbf{y}_i)$ **for** $l = 2$ *to* $L - 1$ **do**
    $\mathbf{f}_i^{(l)} \leftarrow$ Conv+BN+ReLU$(\mathbf{f}_i^{(l-1)})$
  $\hat{\mathbf{n}}_i \leftarrow$ Conv$(\mathbf{f}_i^{(L-1)})$ Update Θ by minimizing $\|\hat{\mathbf{n}}_i - (\mathbf{y}_i - \mathbf{x}_i)\|_2^2$
$\hat{\mathbf{x}} = \mathbf{y} - F(\mathbf{y}; \Theta)$

3 Results and Evaluation

3.1 Direction A: Image Dehazing

3.1.1 Experimental Setup

Datasets To comprehensively evaluate the dehazing algorithms, we utilized three distinct datasets representing different domains:

- **RESIDE-SOTS (Outdoor):** We selected the Synthetic Objective Testing Set (SOTS) outdoor partition, containing 500 distinct hazy images with corresponding ground truth clean images. This dataset serves as our primary benchmark for quantitative analysis using full-reference metrics.
- **O-HAZE:** To assess performance on realistic haze, we employed the O-HAZE dataset, which consists of 45 pairs of outdoor scenes. Unlike synthetic haze, these images contain real haze generated by professional haze machines, providing a more challenging and realistic test bed.
- **Real-world Captures:** We also curated a small set of photographs captured by the team in uncontrolled outdoor environments to test the generalization capability of the algorithms in practical scenarios.

Implementation Details For the **Dark Channel Prior (DCP)**, we implemented the algorithm using Python and OpenCV. To ensure a fair comparison, we conducted experiments in two phases:

1. *Default Setting:* We initially applied standard parameters ($\omega = 0.95, t_0 = 0.1$, patch size 15×15) as described in the original literature.
2. *Parameter Tuning:* Recognizing that standard parameters may not be optimal for all datasets, we performed a parameter search on the RESIDE validation set. We fine-tuned the transmission retention rate (ω), the lower bound (t_0), and the window size. The optimized parameters were determined to be window size $= 31$, $\omega = 0.88$, and $t_0 = 0.15$.

For the learning-based method, we utilized **AOD-Net**. We employed a pre-trained model implemented in PyTorch. The network takes a normalized hazy image as input and outputs the clear image directly. No further fine-tuning was performed on the AOD-Net weights to represent an "off-the-shelf" deep learning baseline.

Evaluation Metrics We employed both full-reference and no-reference metrics to quantify restoration quality:

- **PSNR & SSIM:** For the RESIDE dataset where ground truth is available, we report the Peak Signal-to-Noise Ratio (PSNR) to measure pixel-wise fidelity and the Structural Similarity Index (SSIM) to assess structural preservation.
- **NIQE & BRISQUE:** For the O-HAZE dataset and real-world captures, we utilize the Naturalness Image Quality Evaluator (NIQE) and Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE). Lower scores in these metrics indicate better perceptual quality and fewer natural scene statistic distortions.

3.1.2 Quantitative Results

We first evaluate the restoration fidelity on the synthetic RESIDE-SOTS dataset, followed by a perceptual quality assessment on the real-world O-HAZE dataset.

Performance on Synthetic Data (RESIDE) Table 1 summarizes the PSNR and SSIM performance on the 500 outdoor test images. The results demonstrate the critical importance of parameter tuning for model-based methods.

- **Impact of Tuning:** The default DCP parameters yielded a mean PSNR of 16.27 dB. By optimizing the patch size (31×31) and haze removal factor ($\omega = 0.88$), the performance significantly improved to **20.34 dB**, a gain of over 4 dB.
- **Model-based vs. Learning-based:** Interestingly, the optimized DCP slightly outperformed the AOD-Net baseline (19.55 dB) on this specific dataset. This suggests that for

homogenous synthetic haze models, a well-tuned physical prior can be as effective as a lightweight data-driven model.

Figure 1 visualizes the distribution of PSNR scores across the dataset. While AOD-Net shows a stable distribution, the optimized DCP achieves a higher median and upper quartile, indicating its effectiveness on a majority of the test samples.

Table 1. Quantitative comparison on RESIDE-SOTS (Outdoor) dataset (500 images). Comparison of default DCP, optimized DCP, and AOD-Net. Best results are highlighted in **bold**.

Method	Mean PSNR	Std PSNR	Mean SSIM	Std SSIM
DCP (Default)	16.27 dB	3.23	0.824	0.066
DCP (Tuned)	**20.34 dB**	3.05	**0.900**	0.036
AOD-Net	19.55 dB	2.34	0.893	0.052

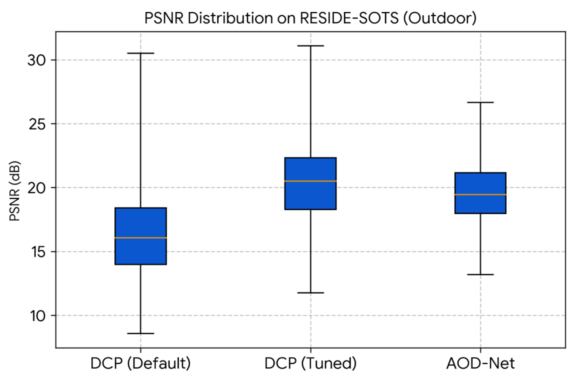

Fig. 1. Distribution of PSNR scores on the RESIDE dataset (outdoor). Parameter tuning (DCP Tuned) significantly shifts the distribution towards higher fidelity compared to the default setting, matching even exceeding the performance of AOD-Net.

Performance on Real-World Data (O-HAZE) Although O-HAZE provides paired clean references, we evaluate it under a no-reference setting to emulate practical deployment scenarios where ground truth is typically unavailable.

Both methods successfully reduce the perceived haze, as evidenced by the improvement over the original hazy images. DCP (Tuned) achieves the lowest BRISQUE score (16.21), suggesting that its strong dark channel constraint effectively restores contrast and natural image statistics. AOD-Net also improves upon the hazy inputs (BRISQUE 17.86) and achieves a competitive NIQE score of 3.72. The larger improvement in BRISQUE for DCP aligns with the visual observation that DCP tends to produce higher contrast results, which often score well on natural scene statistic metrics, though typically at the risk of halo artifacts.

3.1.3 Comparison

The comparative analysis of the Dark Channel Prior (DCP) and AOD-Net reveals distinct performance characteristics that stem from their fundamental methodological differences.

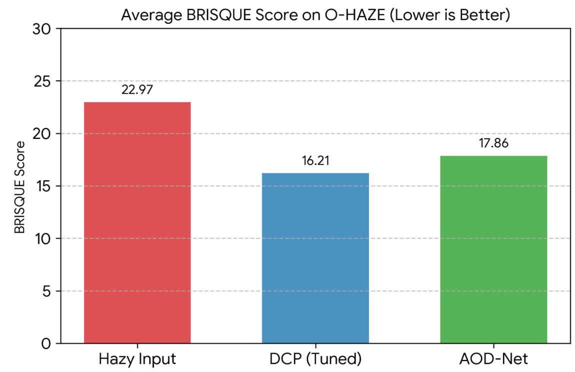

Fig. 2. Comparison of BRISQUE scores on the O-HAZE dataset. Both restoration methods improve the natural scene statistics compared to the hazy input, with tuned DCP showing a slightly stronger reduction in BRISQUE score.

Fig. 3. Visual comparison on synthetic haze (RESIDE). Note that while DCP recovers sharper details, it introduces slight color shifts. AOD-Net preserves the original color tone more faithfully.

Visual Fidelity and Artifacts As illustrated in Figure 3, the two methods exhibit a clear trade-off between local contrast and global coherence.

- **DCP:** Being a physics-based model, DCP excels at removing thick haze and recovering sharp edges, leading to results with strong local contrast. However, its reliance on patch-based statistics introduces specific artifacts. As seen in the magnified regions of Figure 3, "halo" artifacts frequently appear around depth discontinuities (e.g., tree branches against the sky) where the patch-based transmission estimation is inaccurate.
- **AOD-Net:** In contrast, the learning-based AOD-Net produces visually smoother results. It avoids blocky artifacts and maintains better global color fidelity. However, the restored images can sometimes appear "softer" with less aggressive edge definition compared to the tuned DCP.

Fig. 4. Visual comparison on real-world captures. **Left:** Hazy Input. **Middle:** DCP Result. **Right:** AOD-Net Result. DCP tends to oversaturate or darken the sky (a violation of the prior), whereas AOD-Net handles the sky region naturally.

Robustness on Real-World Data The distinction becomes more pronounced in real-world scenarios (Figure 4), where the ideal atmospheric scattering model often holds only approximately.

- **Assumption Violation:** The primary limitation of DCP is its failure in sky regions, where the dark channel assumption ($\min(J) \approx 0$) is invalid. This leads to the sky being misinterpreted as heavy haze, resulting in significant color distortion or darkening (visible in the sky area of the DCP result in Figure 4).
- **Data-Driven Generalization:** AOD-Net demonstrates superior robustness in these cases. By learning from a diverse distribution of images that includes sky and non-uniform haze, it avoids the specific failure modes of analytical priors. It produces a more natural output, albeit with a more conservative dehazing strength.

3.1.4 Discussion

The results highlight a fundamental divergence between physics-based and data-driven dehazing paradigms. The Dark Channel Prior (DCP) benefits from a strong, interpretable physical constraint, allowing it to aggressively recover contrast and visibility in heavy haze. However, its performance is brittle: it relies heavily on hyperparameter tuning (as evidenced by the 4 dB performance gap between default and tuned settings) and struggles in regions where the statistical prior is invalid, such as bright skies.

In contrast, AOD-Net demonstrates the advantage of learned representations. By optimizing an end-to-end reconstruction loss, the network implicitly learns to handle edge cases like sky regions without manual constraints, resulting in higher global color fidelity and fewer artifacts. This suggests that while analytical methods offer interpretability and high sharpness under ideal conditions, deep learning approaches provide superior robustness and consistency for general-purpose restoration, albeit sometimes at the cost of slightly conservative contrast enhancement.

3.2 Direction B: Image Deblurring

3.2.1 Experimental Setup

We evaluated two blind deblurring algorithms using the Sun et al. dataset[18], a widely used benchmark providing pairs of sharp ground-truth images and their synthetically blurred counterparts generated with various motion blur kernels. We selected ten representative images from this dataset, each of which was blurred with eight distinct kernels. This resulted in a total of eighty blurred inputs. We applied both the L0-based method of Xu et al. and the internal patch recurrence method of Michaeli et al. to every blurred image. We assessed the performance of each algorithm against the corresponding sharp ground-truth image.

The experiments were implemented in MATLAB. To ensure fairness, we used the same evaluation pipeline for both algorithms. We set the parameters of each algorithm according to their respective papers whenever possible, making minor adjustments only when required for stability. The evaluation metrics included peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM), both of which were computed over the full-resolution reconstructed image. These two metrics jointly capture pixel-wise fidelity and perceptual similarity, enabling a thorough analysis of deblurring performance. We assume that the blur kernels in the dataset accurately approximate real motion blur, and that PSNR and SSIM across kernels and images provide meaningful aggregate measures of algorithm robustness.

We also tested two blind deblurring algorithms on a small collection of blurred photographs captured in the real world. Because there are no ground-truth images available for these photographs, we could not use evaluation metrics such as PSNR and SSIM. Instead, we used the Naturalness Image Quality Evaluator (NIQE). NIQE is a widely adopted, no-reference image quality metric that quantifies perceptual naturalness based on the statistical regularities of natural images. A lower NIQE score indicates higher perceptual quality. By comparing the NIQE values of the deblurred results produced by the two methods, we could assess their relative performance in real-world conditions where ground truth is unavailable.

3.2.2 Results

The quantitative results are summarized in Figures 5, 6, 7, and 8, which plot the average PSNR and SSIM across images and kernels. The first pair of plots shows how the average PSNR and SSIM vary with respect to the image ID, illustrating the performance of each method across different image contents. The second pair reports the averages over kernels, highlighting how sensitive each algorithm is to different blur types.

Across all metrics, the blurred input consistently has the lowest PSNR and SSIM values, confirming the difficulty of the synthetic blur in the dataset. Xu et al.'s method notably improves upon the blurred baseline, producing reconstructions with PSNR values typically ranging from 26-31 dB and SSIM values around 0.84-0.89. The PSNR of Michaeli et al.'s method surpasses Xu et al.'s for most of the tested images and kernels. However, the SSIM of Michaeli et al.'s method is slightly lower. Its PSNR values are typically in the range of 26–33 dB, and its SSIM values are around 0.81–0.88. Notably, Xu et al.'s method exhibits less variation in performance across images and kernels, suggesting greater robustness.

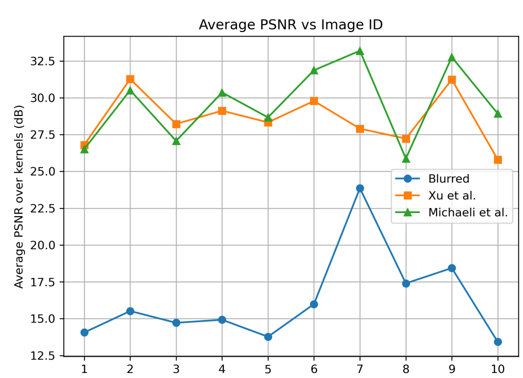

Fig. 5. Average PSNR vs Image ID

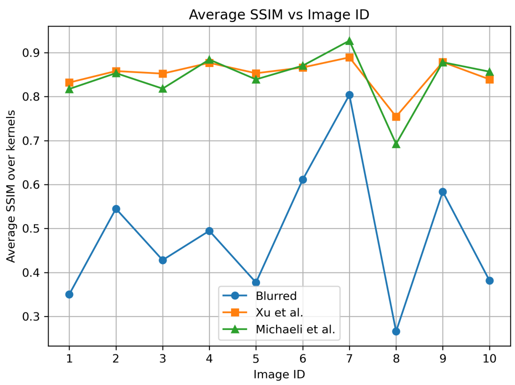

Fig. 6. Average PSNR vs Image ID

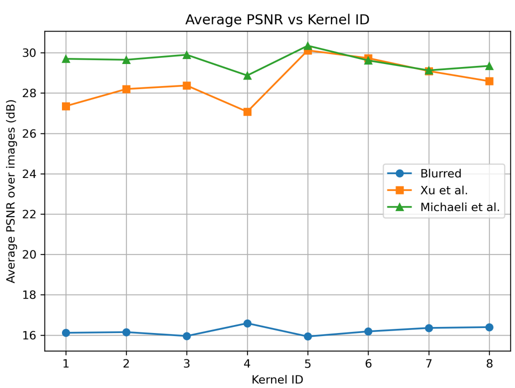

Fig. 7. Average PSNR vs Image ID

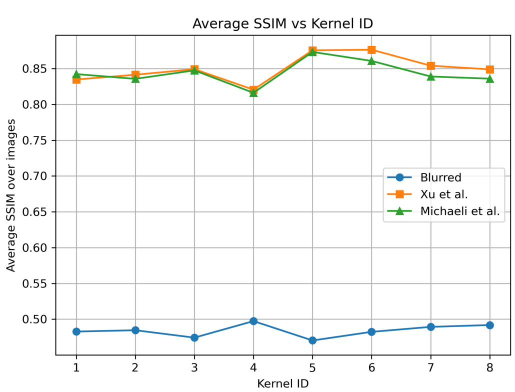

Fig. 8. Average PSNR vs Image ID

In a real-world dataset, The NIQE scores reveal that both algorithms substantially reduce the NIQE values relative to the blurred input. This indicates an improvement in perceptual naturalness. Notably, the second image is unfocused and blurred, while the others are motion blurred. This demonstrates that Xu et al.'s method is more effective in defocus scenarios, whereas Michaeli et al.'s method is more effective in motion blur scenarios.

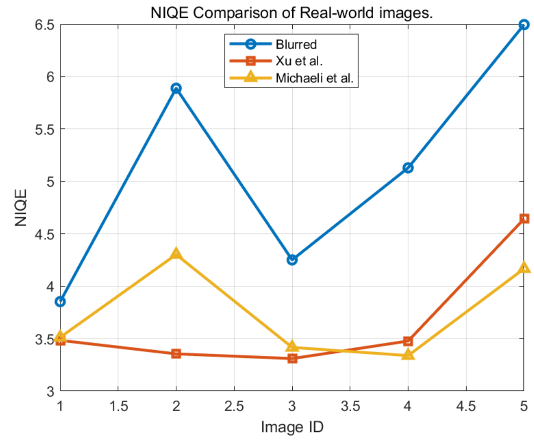

Fig. 9. NIQE Comparison of Real-world images

3.2.3 Comparison

The comparative analysis of two blind deblurring algorithms reveals their distinct performance characteristics, which stem directly from their underlying modeling assumptions. On the synthetic Sun et al. dataset, the method proposed by Michaeli et al. consistently achieves higher PSNR across most image and kernel combinations. Its reliance on cross-scale internal patch recurrence enables more effective restoration of fine structures and textures, especially in scenes containing repetitive patterns or rich local detail. In contrast, Xu et al.'s method emphasizes edge sharpness and is driven by an L0 sparsity prior on image gradients. It tends to produce stable reconstructions with relatively small variance across different blur kernels. This stability is evident in its SSIM scores, which fluctuate less than those of the Michaeli method, despite being slightly lower on average. Thus, Xu et al.'s method appears more robust to variations in blur type, while Michaeli et al.'s method excels in recovering high-frequency details.

A comparison of the two approaches on a real-world dataset further highlights their complementary strengths. Both substantially reduce NIQE scores, confirming that they improve the naturalness of real-world images. However, their behavior differs depending on the type of blur. In an unfocused image where spatial defocus dominates motion blur, for example, Xu et al.'s method produces lower NIQE scores. This suggests that its strong edge-preserving prior is more effective when the blur is spatially smooth. Conversely, for images affected by natural camera shake, Michaeli et al.'s method yields lower NIQE values, indicating superior perceptual quality.

Fig. 10. Real-world Deblurring

3.2.4 Discussion

Overall, the findings suggest that, although both algorithms can perform meaningful blind deblurring, they operate differently and are more effective with certain types of images and blur characteristics. Xu et al.'s method benefits from its mathematically principled L0 gradient minimization framework, which promotes strong sparsity and sharp edge reconstruction. This makes the method reliable and relatively insensitive to kernel variations. It produces stable performance across the synthetic dataset and performs well in real-world images dominated by defocus blur. However, its tendency to oversimplify textures or weaken low-contrast regions affects PSNR and SSIM in scenes with complex details.

In contrast, Michaeli et al.'s method leverages self-similarity within the image across scales. This allows the method to infer missing detail without relying heavily on explicit edge priors. This leads to higher PSNR in nearly all synthetic test cases and better NIQE scores for motion-blurred real-world images. However, since the patch recurrence prior depends on the presence of statistically meaningful self-similar structures, the method is sensitive to image content. It may underperform on unfocused images or scenes with weak internal redundancy, and its SSIM varies more across kernels.

Despite these differences, the two approaches' complementary strengths suggest that blind deblurring performance depends heavily on image characteristics and blur type. These experiments demonstrate the importance of evaluating deblurring algorithms across diverse datasets and of selecting a method that aligns with the expected blur characteristics in practical scenarios.

3.3 Direction C: Image Denoising

3.3.1 Experimental Setup

Three representative denoising methods were compared including Wavelet-based denoising, SVR-based denoising, and DnCNN—under additive Gaussian noise with $\sigma \in \{15, 25, 50\}$. We trained on Zhang's BSD64 dataset, where clean reference images are available and noisy inputs are generated by adding Gaussian noise at each noise level. This enables full-reference quantitative evaluation. In addition, we test all three methods on a small set of real photograph to assess practical behavior under real imaging noise. For the synthetic (paired) dataset, we report PSNR and SSIM. PSNR measures pixel-wise fidelity, while SSIM reflects structural and perceptual similarity. All methods are applied to the same noisy inputs at each σ to ensure fairness.

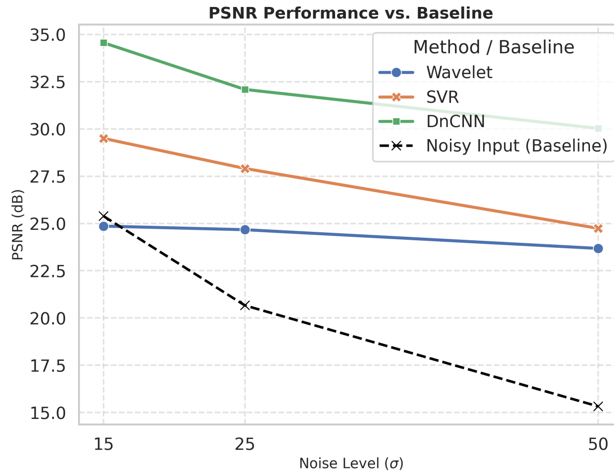

Fig. 11. Denoising performance of PSNR in different noise level

3.3.2 Results

Quantitative Results The quantitative results report the performance of the three denoising methods across noise levels $\sigma \in \{15, 25, 50\}$. DnCNN consistently outperformed both the Wavelet and SVR baselines in terms of PSNR and SSIM. At a low noise level, DnCNN achieved a PSNR of 34.56 dB and an SSIM of 0.901, significantly surpassing SVR (29.50 dB) and Wavelet (24.86 dB). As the noise intensity increased to $\sigma = 50$, the performance gap widened. While the Wavelet method showed limited efficacy with a PSNR improvement of only ≈ 8.3 dB over the noisy input, DnCNN maintained robust performance at 30.03 dB. Notably, although SVR yielded higher PSNR than the Wavelet approach , it exhibited substantially lower structural similarity (0.433 vs. 0.686), suggesting that SVR reduces pixel-wise error at the expense of structural fidelity. Overall, DnCNN demonstrated the highest resilience to severe noise corruption.

Qualitative Results (Real Images). On real noisy photographs, the same ranking is observed visually. Wavelet denoising effectively reduces noise but tends to blur edges and thin structures due to its fixed analytical prior. SVR better preserves some structures but may leave residual noise or introduce slight artifacts when the noise is heavy or non-Gaussian. DnCNN produces the cleanest outputs with the best edge/texture retention in most cases, reflecting the advantage of learned convolutional features in separating noise from image content.

3.3.3 Comparison

As substantiated by the performance metrics in Figure, the proposed DnCNN architecture demonstrates superior denoising performance across the full range of tested noise levels. While traditional approaches exhibit characteristic limitations—the Wavelet transform tends to preserve structural information but delivers only limited gains in signal-to-noise ratio, and SVR achieves moderate noise suppression at the expense of noticeable structural degradation—DnCNN effectively mitigates these trade-offs by simultaneously

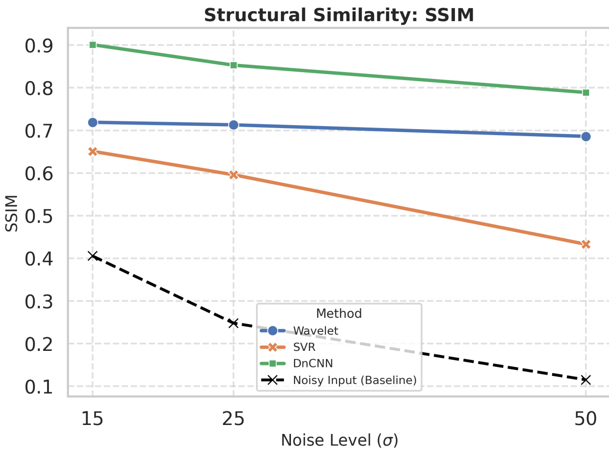

Fig. 12. Denoising performance of SSIM in different noise level

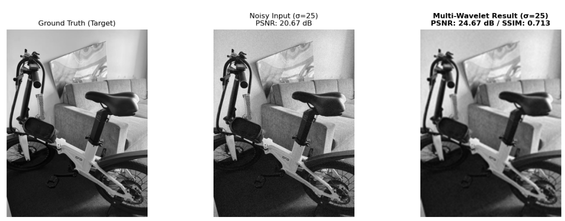

Fig. 13. Wavelet denoising

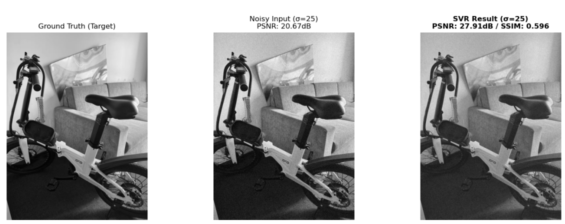

Fig. 14. SVR denoising

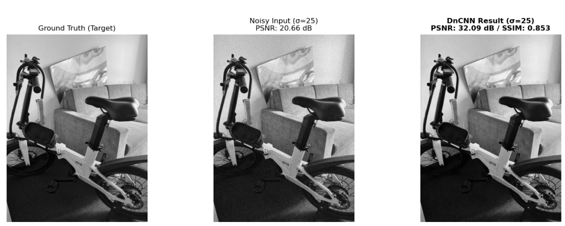

Fig. 15. DnCNN denoising

improving both fidelity and structural integrity. This advantage becomes particularly pronounced under extreme corruption ($\sigma = 50$): while competing methods struggle to recover coherent image features, DnCNN maintains robust performance with PSNR exceeding 30 dB and outperforms the next best method (SVR) by 5.29 dB. This gap highlights DnCNN's capacity to learn complex nonlinear mappings that preserve fine-scale details even when the input is heavily contaminated.

3.3.4 Discussion

The results highlight a fundamental limitation of conventional, non-deep denoising methods: pixel-wise fidelity and perceptual structural integrity are difficult to optimize simultaneously. The wavelet approach acts as a conservative prior-driven filter that suppresses noise while maintaining relatively stable structural consistency, but it tends to offer only limited gains in reconstruction fidelity and may blur fine textures. In contrast, SVR emphasizes reducing pixel-level error through nonlinear regression, which can improve apparent fidelity yet often introduces oversmoothing artifacts that weaken structural similarity. These complementary failure modes suggest that shallow paradigms, constrained by handcrafted representations and limited model capacity, are inherently forced to trade noise removal against the preservation of high-frequency details.

By comparison, DnCNN exhibits a stronger ability to decouple noise from underlying image content without sacrificing structure. Its residual learning formulation and deep convolutional representations enable more effective exploitation of spatial correlations, allowing the network to remove noise while retaining edges and textures. This robustness is especially evident under heavy corruption, where shallow baselines increasingly struggle to preserve coherent structures. Overall, the findings indicate that deep convolutional priors provide a more reliable mechanism for jointly improving fidelity and perceptual quality than classical filtering or shallow regression-based alternatives.

4 Future Work

4.1 Direction A: Image Dehazing

Our comparative study highlighted that while both classical and learning-based methods perform well, there is room for improvement in handling complex real-world conditions. Future work will focus on two main directions.

First, we aim to address the domain gap between synthetic training data and real atmospheric scattering through fine-tuning on real-world data. The current parameters for the Dark Channel Prior (such as patch size and the haze removal factor ω) were fixed based on heuristics. Collecting a dataset of real-world hazy scenes would allow us to statistically determine the optimal parameters for different environmental conditions. Similarly, the learning-based model can be fine-tuned on real images to better preserve global color fidelity and avoid the color shifts occasionally observed in our experiments.

Second, we plan to investigate stronger model architectures. While AOD-Net is computationally efficient, its lightweight CNN structure limits its capacity to handle heavy or non-uniform haze. We intend to explore attention-based dehazing networks (such as Transformers or FFA-Net), which leverage attention mechanisms to better capture long-range dependencies and spatially varying haze features, potentially yielding superior restoration quality than standard convolutional networks.

4.2 Direction B: Image Deblurring

There are several areas that remain for future exploration. The two algorithms evaluated in this study perform well on both synthetic and real-world blur; however, their underlying assumptions limit their generality. Evaluating them on more diverse blur types, such as spatially varying motion, rolling shutter distortions, and low-light noise, would provide a clearer picture of their practical robustness. Another promising area for improvement is computational efficiency, particularly for the patch-based method of Michaeli et al., which is relatively costly. Finally, recent advances in deep learning offer opportunities to integrate classical prior interpretability, such as gradient sparsity or internal patch recurrence, into learning-based frameworks. This could combine the strengths of traditional and modern approaches.

4.3 Direction C: Image Denoising

While the proposed DnCNN model demonstrates robust performance on synthetic additive white Gaussian noise (AWGN), real-world imaging scenarios often involve complex, signal-dependent noise distributions. Consequently, future research will focus on extending this framework to blind image denoising on real-world datasets, such as the Smartphone Image Denoising Dataset (SIDD). Additionally, we aim to investigate lightweight architecture designs—employing techniques such as model pruning or quantization—to reduce computational overhead, thereby facilitating the deployment of deep denoising models on resource-constrained edge devices.

5 Conclusion

In conclusion, this project establishes a unified experimental framework to compare classical and learning-based image restoration methods across dehazing, deblurring, and denoising, providing systematic evaluations on public dataset and real-world campus images to analyze the strengths and limitations of analytical priors versus learned representations under realistic acquisition conditions.

For dehazing, we implemented a classical Dark Channel Prior (DCP) baseline and a lightweight deep-learning baseline (AOD-Net). On synthetic benchmarks with paired ground truth, both methods substantially improved PSNR/SSIM over hazy inputs, and tuning DCP parameters further boosted its performance. On real-world hazy images without ground truth (O-HAZE and our self-captured photos), both methods reduced no-reference quality scores (NIQE/BRISQUE), with DCP exhibiting slightly better robustness under domain shift while AOD-Net produced smoother color transitions. Overall, our results highlight a trade-off between physics-based priors and data-driven models: learning-based methods can achieve strong restoration on data distributions similar to training, whereas classical priors often generalize more reliably to real-world haze.

For deblurring, we evaluated L0-based deblurring method of Xu *et al.* and the internal patch recurrence method of Michaeli *et al.*. Both methods produce meaningful improvements over the blurred inputs, but they exhibit different strengths that stem directly from their underlying priors. The L0-based approach tends to generate stable and visually sharp reconstructions, demonstrating strong robustness across a wide range of blur cases, while the internal patch recurrence method is often more effective at recovering fine textures and repetitive structures when sufficient self-similarity exists in the image. On real photographs, both methods improve perceptual naturalness, yet their relative advantages vary with blur characteristics, indicating that practical deblurring performance is highly content- and blur-dependent. These findings suggest that no single classical prior dominates across all scenarios; instead, selecting an algorithm should be guided by the expected blur type and image content.

For denoising, we implemented three methods of multi-wavelet shrinkage, support-vector regression (SVR), and DnCNN. The quantitative evaluation conclusively demonstrates that the proposed DnCNN architecture overcomes the inherent performance trade-offs observed in traditional baselines. While the Wavelet transform and SVR methods exhibited distinct limitations in balancing noise suppression with structural preservation—particularly under high-noise regimes—DnCNN consistently achieved superior fidelity across all metrics. Notably, under severe noise corruption ($\sigma = 50$), the proposed method demonstrated exceptional robustness, outperforming the SVR baseline by a significant margin of **5.29 dB**. These findings confirm that leveraging deep residual priors allows for the effective recovery of latent image features from heavily corrupted observations.

Across all experiments, employing consistent PSNR, SSIM, and NIQE metrics enabled a fair comparison of reconstruction fidelity, perceptual quality, and computational efficiency in campus imaging scenarios. Overall, the results suggest that classical methods remain practical and reliable when training data or compute budgets are limited, whereas deep networks deliver performance advantages when sufficient data are available for effective learning and generalization.

6 Project Repository

Our implementation, instructions, and reproducibility materials are available on GitHub: `https://github.com/chakogithub/ECE253-Final`. The repository includes a README with step-by-step commands to run all experiments.

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